Inferring Subject-Specific Distortion Fields from High-Dimensional Neural Color Representations

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Abstract

Color vision deficiency (CVD) distorts the continuous geometry of cortical color representations, but characterizing this distortion from high-dimensional fMRI signals and mapping it to low-dimensional mechanistic parameters remains challenging. We introduce a framework for fitting parsimonious, retina-inspired distortion models (1–2 free parameters) to individual fMRI-measured hue interpolation profiles in visual cortex. Applied to two CVD and one control participant, the framework reveals that each CVD case is best captured by a *different* model—a 1-parameter retinal cone-shift for moderate protanomaly and a 2-parameter retinal–cortical model for mild deuteranomaly—while the control is correctly rejected by all models. A complementary cortical-level analysis recovers an S-cone compensatory rotation of $\sim 21.5°$, converging with the $21.4°$ independently reported from behavioral hue-scaling, and provides cross-criterion validation linking per-color interpolation accuracy to pairwise representational geometry. The fitted models further predict whether stimulus-space correction is feasible: the degree of geometric collapse in the distortion field determines correction limits.

1. Introduction

Color vision deficiency (CVD) arises from altered spectral sensitivity of retinal cone photoreceptors and affects approximately 8% of males worldwide (Machado et al., 2009). Current assistive approaches, including commercial spectral-notch filters, apply a fixed, generic correction. Recent evidence indicates that such filters can alter color *appearance* but have minimal effect on color *discrimination* at threshold (Somers et al., 2024), suggesting that effective correction may require individually-tailored strategies informed by each person's neural color geometry.

A central challenge is that CVD-related neural distortion is not uniform across individuals. Even within a single diagnostic category (e.g., deuteranomaly), the degree of cortical compensation varies substantially (Tregillus et al., 2021), and the functional deficit that matters for continuous color tasks—the ability to interpolate between adjacent hues, not merely classify them—differs in ways that categorical diagnosis cannot capture. Characterizing this distortion requires moving from classification-based neural decoding to *geometry-sensitive* evaluation of high-dimensional fMRI representations: asking not whether the brain can tell two colors apart, but whether the continuous structure between them is preserved.

Here we propose a framework for inferring subject-specific distortion fields from fMRI data, casting CVD as a case study in recovering low-dimensional distortion structure from complex representational geometry. We fit parsimonious mechanistic forward models to each CVD participant's neural interpolation profile, as measured by a geometry-sensitive leave-one-color-out (LOCO) cross-validation metric. Our contributions are:

1. We identify subject-specific interpolation failure as a geometry-sensitive phenotype that dissociates from classification accuracy, with the effect most robustly captured in hV4.

2. We introduce a personalized distortion-fitting framework in which different CVD participants are best captured by different parsimonious models (1–2 DOF), with successful specificity rejection of a normal control.

3. We demonstrate that correction feasibility depends on the degree of geometric collapse induced by the fitted forward model, revealing severity-dependent limits on exact stimulus-space intervention.

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2. Methods

2.1. Participants and Stimuli

Ten participants were included in the analysis (7 healthy controls [HC], 3 females, age 22.7 ± 2.5 years; 3 color vision deficient [CVD], all male, including one deuteranope, one protanomalous, and one near-normal deutan serving as a specificity control). Color vision was assessed using the 14-plate Ishihara test (Ishihara, 1917). Given the small CVD sample, all CVD results are interpreted as individual case demonstrations, not population-level effects (Crawford & Howell, 1998).

Stimuli consisted of eight colors equally spaced from 0° to 315° in CIE $L^*a^*b^*$ space, with lightness fixed at $L^* = 75$ and chroma distance of 40 from the achromatic point. Participants performed a modified Rapid Serial Visual Presentation (RSVP) detection task (Brouwer & Heeger, 2009) across six runs per session, maintaining fixation while viewing 1.5 s color patches presented in optimized order. Functional images were acquired on a Siemens 3T scanner (TR = 1.5 s, 2 mm isotropic voxels, 24 oblique slices perpendicular to the calcarine sulcus), preprocessed with fMRIPrep, and projected to retinotopically-defined ROIs (V1, V2, hV4) using the Wang probabilistic atlas at 50% threshold (Wang et al., 2015). Response amplitudes were estimated using a two-stage FIR-then-GLM procedure (Brouwer & Heeger, 2009) and Procrustes-aligned across runs within each participant to improve cross-run consistency.

2.2. Neural Phenotype: Interpolation Vulnerability

We modeled hue-selective neural populations using half-wave rectified and squared sinusoidal basis functions (Brouwer & Heeger, 2009): each of K hypothetical channels was tuned to a preferred hue θ_k, with response $r_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$. The weight matrix W mapping channel outputs to voxel responses was estimated via ridge regression with generalized cross-validation (GCV) for regularization (Golub et al., 1979).

The target phenotype is the *LOCO vulnerability profile*, a geometry-sensitive metric of representational structure. Leave-one-color-out (LOCO) cross-validation tests whether the model can interpolate to novel hues not seen during training, assessing the preservation of continuous color-space geometry in the high-dimensional voxel pattern space: W is trained on 7 of 8 colors and voxel responses to the held-out color are predicted. The per-color prediction error defines an 8-element vulnerability vector $\mathbf{v} \in \mathbb{R}^8$ for each participant, capturing which hues are poorly interpolated from their neighbors—a structural deficit invisible to classification-based decoders.

This phenotype is specific to continuous geometry and distinct from color discrimination. Leave-one-*run*-out (LORO) cross-validation, which tests whether individual colors can be reliably distinguished across repetitions, shows no HC–CVD difference ($p = 0.668$), while LOCO interpolation error is significantly elevated in CVD. Among the ROIs tested, the effect was most robust in hV4 ($p = 0.017$; K=3), consistent with hV4's established role in perceptual color representation (Brouwer & Heeger, 2009; Bannert & Bartels, 2018). We therefore focus distortion fitting on hV4, with V1 analyses reported as cross-ROI validation in Section A.

2.3. Distortion Models

We fit two primary forward models of hue distortion (Figure 1a), each parameterizing how a physical stimulus angle θ is transformed into a distorted angle θ':

Machado cone shift (1 DOF). A physiological model (Machado et al., 2009) in which the anomalous cone's spectral peak is shifted by $\Delta\lambda \in [0, 20]$ nm. This retinal-level model predicts hue distortion through altered L–M cone opponency.

Retinal + cortical gain (R+C; 2 DOF). Extends the cone shift with a cortical opponent-channel gain g:

$$rg' = rg_{\text{base}} + (1+g) \cdot (rg_{\text{ret}} - rg_{\text{base}}) \tag{1}$$

where g=0 reduces to pure Machado, g=-1 is exact cortical compensation, and g<-1 is overcompensation (Tregillus et al., 2021).

2-component angular dilation (2 DOF; complementary). As a complementary cortical-level analysis, we also fit a model that parameterizes hue distortion as angular dilation along two opponent axes in hue space:

$$\theta'(c) = \theta(c) + \beta_s \cos(\theta(c) - 90^\circ) + \beta_c \cos(\theta(c) - \theta_{\text{conf}}) \tag{2}$$

where β_s captures S-cone axis expansion and β_c captures confusion-axis modulation ($\theta_{\text{conf}} = 16^\circ$ for protan, 150° for deutan). Unlike the primary cone-level models, this operates directly in opponent hue space without modeling retinal spectral shifts, and is additionally validated against pairwise representational distance matrices (ΔRDM) in V1.

2.4. Fitting and Specificity

For each model and candidate parameter(s), we simulate the LOCO vulnerability profile that a healthy observer would exhibit if viewing stimuli distorted by the model's forward map. The fit is the Spearman rank correlation ρ between simulated and observed vulnerability vectors, optimized over a grid ($\Delta\lambda$: 41 points; g: 25 points; β_s, β_c: $26 \times 51 =$ 1,326 points).

Table 1. Subject-specific distortion model fits (hV4 LOCO). Each CVD subject is best explained by a different parsimonious model. The normal control is correctly rejected by all models.

Subject	Best model	DOF	ρ	p_{perm}
Sub-08 (deutan)	R+C	2	0.857	**0.005****
Sub-09 (protan)	Machado	1	0.762	**0.018***
Sub-10 (control)	—	—	−0.048	0.559

Statistical significance is assessed by exact permutation of the 8-color labels ($8! = 40{,}320$ permutations). A valid model must satisfy two criteria:

1. **Sensitivity**: significant fit ($p < 0.05$) for CVD participants.
2. **Specificity**: non-significant fit for the normal control (sub-10).

A model that fits CVD subjects but also yields a false positive on the control is rejected as overfitting.

3. Results

3.1. Subject-Specific Model Selection

Table 1 and Figure 1 present the central result: distinct CVD participants require distinct low-dimensional distortion models, whereas the normal control yields no significant fitted distortion.

Sub-09 (moderate protanomaly). The 1-DOF Machado model with $\Delta\lambda = 13.5$ nm—within the established range for moderate–severe protanomaly (Machado et al., 2009)—captures the interpolation profile at $\rho = 0.762$, $p = 0.018$. Adding a cortical gain parameter does not improve fit ($g = 0.0$ at optimum), indicating that a purely retinal model is sufficient for this participant.

Sub-08 (mild deuteranomaly). The 1-DOF Machado model is only trend-level ($p = 0.058$). The 2-DOF R+C model, with a small retinal shift ($\Delta\lambda = 2.0$ nm) and a cortical gain term ($g = +2.25$), achieves $\rho = 0.857$, $p = 0.005$. The additional parameter is essential: the retinal shift alone explains little of the distortion in this participant's hV4 representation.

Sub-10 (normal control). All cone-level models yield $\Delta\lambda \approx 0$ and non-significant fits ($p = 0.559$), confirming specificity.

3.2. Fitted Parameters

The fitted retinal parameters fall within established physiological ranges: sub-08's $\Delta\lambda = 2.0$ nm matches very mild deuteranomaly, and sub-09's $\Delta\lambda = 13.5$ nm matches moderate protanomaly (Machado et al., 2009). Sub-09's cortical

[FIGURE PLACEHOLDER]
Figure 1: Subject-Specific Distortion Profiles

(a) Model schematic: stimulus $\theta \rightarrow$ retinal cone shift ($\Delta\lambda$) $\rightarrow$ optional cortical gain (g) $\rightarrow$ distorted θ' $\rightarrow$ forward encoding $\rightarrow$ LOCO evaluation.

(b–d) 8-color LOCO vulnerability profiles. Gray bars: observed CVD profile. Colored line: best-model fit. **(b)** Sub-08: R+C fit (ρ=0.857, p=0.005). **(c)** Sub-09: Machado fit (ρ=0.762, p=0.018). **(d)** Sub-10: flat profile, p=0.559 (specificity pass).

Figure 1. The central result. Each CVD participant exhibits a distinct vulnerability profile (gray bars) in the high-dimensional voxel representation, captured by a different parsimonious model (colored lines). The normal control shows no structured vulnerability. Panel (a) illustrates the model hierarchy.

gain collapses to $g = 0$, consistent with a purely retinal account.

Sub-08's fitted cortical gain ($g = +2.25$) is less straightforward to interpret. One possibility is that the mild retinal deficit ($\Delta\lambda = 2.0$ nm) produces a small spectral signal that cortical processing amplifies into a measurable interpolation deficit in hV4—an effect invisible under retinal-only modeling. However, g could alternatively be absorbing unmodeled variance (e.g., individual differences in voxel noise or attentional modulation) rather than reflecting a genuine cortical mechanism. We treat this parameter as a provisional effective description of sub-08's distortion, not as a direct physiological measurement. Cortical compensation magnitudes are known to vary widely across individuals (Tregillus et al., 2021), but the specific sign and magnitude observed here ($g > 0$, amplification rather than compensation) lacks direct precedent and requires independent validation.

3.3. Complementary Cortical-Level Analysis: S-Cone Compensation

The 2-component angular dilation model (Equation (2)) provides a complementary perspective by operating directly in opponent hue space rather than modeling retinal spectral shifts. Under the LOCO criterion, this model reaches significance for both CVD participants (sub-08 hV4 $p = 0.004$; sub-09 hV4 $p = 0.035$) but with marginal specificity for the control (sub-10 $p = 0.058$), limiting its role as a standalone primary model.

Its value lies in cross-criterion validation and biological interpretability. When fitted independently to pairwise representational distance matrices (ΔRDM) in V1—a different criterion (geometric distance structure), brain region, and fitting objective—the S-cone expansion parameter yields

$\beta_s = 20.0° \pm 8.0°$ (sub-08) and $\beta_s = 23.0° \pm 10.2°$ (sub-09), with bootstrap confidence intervals excluding zero in both cases. The cross-subject mean $\beta_s \approx 21.5°$ matches the 21.4° B-Y axis rotation independently measured via behavioral hue-scaling (Emery et al., 2021)—convergent evidence from an entirely different method. This convergence suggests that both CVD subtypes share a compensatory S-cone pathway upregulation, consistent with hypothesized long-term neural adaptation (Tregillus et al., 2021; Boehm et al., 2014).

The confusion-axis parameter β_c shows family-appropriate behavior: significantly negative for the deutan ($-18° \pm 6°$, CI excluding 0) and non-significant for the protan ($+3° \pm 2°$, CI including 0), consistent with the larger retinal shift in protanomaly already capturing confusion-axis structure.

This complementary analysis confirms that the distortion captured by the primary cone-level models in hV4 is not criterion- or ROI-specific: it is also visible as geometric distortion in V1 pairwise distance structure, with a biologically interpretable parameterization that converges with independent behavioral data.

3.4. Geometric Collapse Determines Correction Limits

Given a fitted forward model $D\colon \theta \to \theta'$, we ask whether an inverse correction exists—i.e., whether modified stimulus angles θ^* can be found such that $D(\theta^*) = \theta_{\text{target}}$ for all 8 colors.

Sub-08 (mild, $\Delta\lambda = 2.0$ nm): The R+C forward map preserves an opponent arc wide enough that all 8 hues remain separable. Numerical pre-image search yields exact solutions ($< 0.001°$ residual), producing an 8-point stimulus correction lookup table (Figure 2a).

Sub-09 (moderate, $\Delta\lambda = 13.5$ nm): The Machado forward map compresses the 360° opponent arc to approximately 96°, merging green, cyan, and blue into a single perceived angle ($\sim$282°). Exact inversion is structurally impossible for 4/8 colors (Figure 2b). A discriminability-oriented fallback (maximizing minimum pairwise angular separation) achieves 5.6× improvement over the unfiltered state but reaches only 48% of the healthy-observer separation.

This severity-dependent boundary gives the forward model a practical role: the degree of geometric collapse in the fitted distortion field predicts whether exact stimulus-space correction is feasible for a given individual, before behavioral testing is conducted. We note that the complementary cortical-level model, which operates via angular dilation rather than spectral shift, preserves bijectivity by construction and admits exact pre-image solutions for both CVD participants (see Section A). This highlights that correction feasibility is jointly determined by severity and modeling level—a consideration for future intervention design.

[FIGURE PLACEHOLDER]
Figure 2: Geometric Collapse and Correction Limits
(a) Sub-08 (mild): wide opponent arc; pre-image inversion restores all 8 colors ($< 0.001°$ error).
(b) Sub-09 (moderate): arc compresses $360° \to \sim 96°$; three colors merge at $\sim 282°$. Exact inversion impossible. Separation-optimized fallback: 5.6× improvement, 48% of healthy separation.

Figure 2. The degree of geometric collapse in the fitted distortion field determines whether exact stimulus-space correction is achievable. Mild distortion preserves separability and admits exact inversion; moderate distortion collapses multiple hues onto one perceived angle, precluding exact recovery.

4. Discussion

Inferring distortion from complex signals. The core methodological contribution is that subject-specific, low-dimensional distortion fields (1–2 parameters) can be inferred from high-dimensional fMRI voxel patterns (hundreds of voxels × 8 colors × 6 runs). Although we do not perform multimodal fusion directly, the resulting distortion fields offer an interpretable intermediate representation that could support future integration of brain data with other structured health modalities. The LOCO vulnerability profile serves as the bridge: it compresses the high-dimensional representation into a geometry-sensitive 8-element summary that is sensitive to continuous structure rather than categorical boundaries. This approach may generalize beyond CVD to other conditions where sensory representations are distorted rather than absent.

What this framework does and does not establish. We have shown that subject-specific interpolation profiles can be recovered by parsimonious forward models, with correct specificity rejection and physiologically plausible retinal parameters. A complementary cortical-level analysis recovers an S-cone compensatory rotation that converges with independent behavioral measurements ($\beta_s \approx 21.5°$ vs. the 21.4° reported by Emery et al. 2021), providing cross-criterion and cross-ROI validation. We have *not* shown that the resulting corrections improve perceptual discrimination—that requires behavioral validation, which is future work.

Why different models for different subjects? Sub-09's profile is captured by pure cone shift ($g = 0$) while sub-08 requires the additional cortical gain term. This heterogeneity is itself an important result: it suggests that the neural consequences of CVD cannot be reduced to diagnostic category alone, but instead reflect person-specific variation in retinal severity and downstream processing that is only visible under individualized modeling.

Limitations. Three CVD participants are insufficient for population-level claims; this work is a proof-of-concept for individualized distortion modeling. The fMRI requirement limits near-term scalability. The complementary 2-component model achieves marginal specificity on the control participant ($p = 0.058$), warranting caution about overfitting with 1,326 grid points on 8 colors; the ΔRDM validation in V1 provides a partial independent check. The hV4 focus, while motivated by the strongest geometry-sensitive HC–CVD separation under LOCO, means that distortion structure in earlier visual areas is not directly modeled in the main analysis (see Section A).

Relation to prior work. Our approach extends Brouwer & Heeger's (Brouwer & Heeger, 2009) forward encoding framework—which demonstrated LOCO reconstruction in hV4 for healthy observers—to CVD with individual-level distortion fitting. The distinction between preserved classification and impaired interpolation is consistent with reports that CVD individuals maintain categorical but not continuous color representations (Bannert & Bartels, 2018). The cortical compensation parameters relate to accumulating evidence for long-term neural adaptation in anomalous trichromacy (Tregillus et al., 2021; Emery et al., 2021).

5. Conclusion

We present a proof-of-concept framework for inferring subject-specific distortion fields from high-dimensional neural representations in color vision deficiency. The framework extracts a geometry-sensitive interpolation phenotype from fMRI data, fits it with parsimonious mechanistic models, and predicts whether stimulus-space correction is feasible based on the degree of geometric collapse. A complementary cortical-level analysis recovers a compensatory S-cone rotation converging with independent behavioral measurements, providing cross-criterion validation of the distortion structure. The critical next step is behavioral validation: determining whether the model-prescribed corrections actually improve color discrimination.

Impact Statement

This work develops personalized neural models of color vision deficiency with the goal of informing assistive technology design. Potential positive impacts include individually-tailored color corrections for digital displays and accessibility tools. The framework currently requires fMRI data, limiting immediate clinical deployment. Behavioral validation and informed consent are essential prerequisites before any correction is applied, as a misspecified model could impair rather than improve visual function.

References

Bannert, M. M. and Bartels, A. Human V4 activity patterns predict behavioral performance in imagery of object color. *Journal of Neuroscience*, 38(15):3657–3668, 2018.

Boehm, A. E., MacLeod, D. I. A., and Bosten, J. M. Compensation for red-green contrast loss in anomalous trichromats. *Journal of Vision*, 14(13):19, 2014.

Brouwer, G. J. and Heeger, D. J. Decoding and reconstructing color from responses in human visual cortex. *Journal of Neuroscience*, 29(44):13992–14003, 2009.

Crawford, J. R. and Howell, D. C. Comparing an individual's test score against norms derived from small samples. *The Clinical Neuropsychologist*, 12(4):482–486, 1998.

Emery, K. J., Volbrecht, V. J., Peterzell, D. H., and Webster, M. A. Variations in normal color vision. VII. relationships between color naming and hue scaling. *Vision Research*, 183:1–12, 2021.

Golub, G. H., Heath, M., and Wahba, G. Generalized cross-validation as a method for choosing a good ridge parameter. *Technometrics*, 21(2):215–223, 1979.

Ishihara, S. *Tests for Colour-Blindness*. Handaya, Tokyo, 1917.

Machado, G. M., Oliveira, M. M., and Fernandes, L. A. F. A physiologically-based model for simulation of color vision deficiency. *IEEE Transactions on Visualization and Computer Graphics*, 15(6):1291–1298, 2009.

Somers, L. P., Franklin, A., and Bosten, J. M. Empirical tests of the effectiveness of EnChroma multi-notch filters for enhancing color vision in deuteranomaly. *Vision Research*, 218:108390, 2024. doi: 10.1016/j.visres.2024.108390.

Tregillus, K. E. M., Isherwood, Z. J., Vanston, J. E., Engel, S. A., and MacLeod, D. I. A. Color compensation in anomalous trichromats assessed with fMRI. *Current Biology*, 31(5):936–942, 2021.

Wang, L., Mruczek, R. E. B., Arcaro, M. J., Bhatt, S., and Kastner, S. Probabilistic maps of visual topography in human cortex. *Cerebral Cortex*, 25(10):3911–3931, 2015.

A. Supporting Analyses

A.1. Cross-ROI and Cross-Criterion Evidence

Beyond the primary hV4 LOCO fitting, the 2-component model was validated across multiple ROIs and criteria. Sub-08 V1 LOCO reached $p = 0.001$ ($\beta_s = 50°$, $\beta_c = -14°$)—the strongest LOCO result in the pipeline. Sub-09 V1 ΔRDM reached $p = 0.007$ (crossnobis cosine). The 2-component model is the only model achieving significance under both LOCO and ΔRDM criteria for both CVD participants (Table 2).

A.2. Model-Dependent Correction Feasibility

Under the cortical-level 2-component model, both CVD participants admit exact pre-image solutions for all 8 colors (sub-08: mean $|\delta| = 46.3°$; sub-09: mean $|\delta| = 20.1°$; residual $< 0.001°$). This contrasts with the retinal Machado model, where sub-09's arc compression makes exact inversion impossible for 4/8 colors. The difference arises because angular dilation in opponent hue space preserves bijectivity by construction, whereas retinal spectral shifts can compress the opponent arc irreversibly.

For sub-08, the R+C and 2-component models prescribe substantially different correction directions (cosine between correction vectors = -0.18; sign agreement 3/8), despite comparable fit quality (R+C $\rho = 0.857$ vs. 2-component $\rho = 0.881$). Which correction actually improves perception requires behavioral validation.

A.3. Model Comparison: Full Table

Table 2. Full model comparison across hV4 LOCO, V1 LOCO, and V1 ΔRDM criteria. The 2-component model is the only model achieving significance under both LOCO and ΔRDM for both CVD participants.

Subject	Model	DOF	hV4 LOCO ρ	hV4 LOCO p	V1 LOCO p	V1 ΔRDM	Notes
Sub-08	Machado	1	0.619	0.058	0.033*	anti-corr.	Trending (hV4)
	R+C	2	**0.857**	**0.005****	0.047*	0.179	hV4 primary
	2-Comp	2	0.881	0.004**	**0.001*****	CI excl. 0	Cross-criterion
Sub-09	Machado	1	**0.762**	**0.018***	0.112	+0.091	hV4 primary
	R+C	2	0.762	0.018	—	0.026*	g=0 (= Machado)
	2-Comp	2	0.690	0.035*	0.018*	**0.007*****	Cross-criterion
Sub-10	Cone-level	—	−0.048	0.559	—	—	Specificity pass
	2-Comp	2	—	0.058†	—	—	Marginal

† Marginal specificity; see Limitations discussion.

A.4. Correction Feasibility Details

Sub-08 (R+C pre-image). All 8 colors admit exact pre-image solutions (mean $|\delta| = 23.5°$, max $= 42.9°$, residual $< 0.001°$). The resulting 8-point lookup table is directly implementable on any digital display.

Sub-09 (Machado pre-image). At $\Delta\lambda = 13.5$ nm, the opponent-space arc compresses from 360° to ∼96°. Colors c4 (green), c5 (cyan), and c6 (blue) converge to a single perceived angle (∼282°). Exact pre-image: 4/8 colors. Separation-optimized fallback: min separation $1.03° \rightarrow 5.76°$ ($5.6\times$), mean separation $39.3° \rightarrow 24.3°$ (healthy: 70.7°).

Sub-09 (2-component pre-image). All 8 colors admit exact pre-image solutions (mean $|\delta| = 20.1°$, max $= 48.1°$, residual $< 0.001°$). The cortical-level angular dilation preserves bijectivity, unlike retinal spectral shift, enabling exact correction where the Machado model predicts irreversible collapse.